

AI-Powered “Bail Reckoner”

An Intelligent Legal Assistance System for Streamlining Bail Eligibility and Application

Extended Project Abstract · Major Project (Stage-I) · Academic Year 2026–27

Field	Detail
Problem Statement ID	SIH1702 — “Bail Reckoner”
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Institute	Marri Laxman Reddy Institute of Technology and Management, Hyderabad, Telangana

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1. Introduction & Problem Statement

India’s prison system is, in numerical terms, a system that holds people who have not been convicted of anything. According to the National Crime Records Bureau’s *Prison Statistics India 2024*, as on 31 December 2024 the country’s 1,333 jails held **5,11,542** inmates against a sanctioned capacity of **4,53,769** — an occupancy rate of **112.7%**, a decade low, but achieved against a population of which **3,71,440 (about 72.6%) were undertrial prisoners**. Delhi recorded the highest occupancy in the country at 194.6%, followed by Meghalaya at 163.5%. Some 9,028 undertrials — 2.4% of the undertrial population — had been in custody for more than five years without a conviction. [19][20]

Parliament has already legislated a partial answer. **Section 479 of the Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023**, which replaced Section 436A of the Code of Criminal Procedure, 1973 with effect from 1 July 2024, entitles an undertrial who has undergone detention for one-half of the maximum prescribed sentence to be released on bail — and, for a first-time offender, at one-third, on bond. Section 479(3) goes further: it places the duty to move the court not on the prisoner but on the **Superintendent of the jail**, who “shall forthwith make an application in writing to the Court” once the threshold is crossed. [1][2]

The gap between that entitlement and its realisation is the problem this project addresses, and it is measurable. Following a Ministry of Home Affairs advisory dated 24 October 2024 and a Constitution Day release campaign, the Government reported that as on 26 November 2024, across all States and Union Territories, **951 prisoners had been identified as eligible under Section 479 and 334 had been granted bail** — against an undertrial population of roughly 3.71 lakh. [15][16][17] The Supreme Court, hearing *In Re: Inhuman Conditions in 1382 Prisons*, has repeatedly directed States and Union Territories to take proactive steps to identify and release eligible undertrials. [13] [14] The bottleneck is not judicial refusal. It is that nobody has done the arithmetic.

That arithmetic is not trivial. A single bail-entitlement determination requires the analyst to identify every offence charged; determine the governing statute book — IPC 1860 and CrPC 1973 for offences committed before 1 July 2024, or BNS, BNSS and BSA 2023 for offences thereafter, since the new codes operate prospectively and both regimes therefore run in parallel across Indian courts today; check whether any special statute with a stringent bail bar applies (NDPS, PMLA, UAPA, POCSO, the SC/ST Act, the IT Act); retrieve the maximum prescribed sentence for each offence; measure custody undergone from the correct date; apply the one-half or one-third fraction; and — critically — test whether Section 479(2) bars the benefit altogether because more than one offence or multiple cases are pending. Today a legal-aid lawyer, a paralegal volunteer, a jail superintendent or an undertrial’s family performs this by hand, across fragmented sources, often without a lawyer at all.

The **Bail Reckoner** is a system that performs this computation, shows its working, and cites the provision behind every step. It does not predict whether bail will be granted, and it does not model judicial discretion. It computes what the statute already provides and makes that visible to the people whose job it is to act on it.

2. The Statutory Framework the System Encodes

An eligibility engine is only as good as the law it encodes. This section states the provisions the rules engine implements, so that the logic is auditable against the bare act rather than against a paraphrase.

2.1 Section 479(1) BNSS and its three provisos

“Where a person has, during the period of investigation, inquiry or trial under this Sanhita of an offence under any law (not being an offence for which the punishment of death or life imprisonment has been specified as one of the punishments under that law) undergone detention for a period extending up to one-half of the maximum period of imprisonment specified for that offence under that law, he shall be released by the Court on bail.” — Section 479(1), BNSS 2023

- **First proviso** — a *first-time offender* (one who “has never been convicted of any offence in the past”) shall be released *on bond* on completing **one-third** of the maximum period. This route has no predecessor in Section 436A CrPC and remains materially under-invoked.
- **Second proviso** — the Court may, after hearing the Public Prosecutor and *for reasons to be recorded in writing*, order continued detention beyond one-half, or release the person on bail bond instead of bond.
- **Third proviso** — in no case shall a person be detained during investigation, inquiry or trial for longer than the maximum period of imprisonment prescribed for the offence.
- **Exclusion in the main clause** — the sub-section does not apply where death or life imprisonment is one of the prescribed punishments.

2.2 Section 479(2) — the exclusion that governs most real cases

“Notwithstanding anything in sub-section (1), and subject to the third proviso thereof, where an investigation, inquiry or trial in more than one offence or in multiple cases are pending against a person, he shall not be released on bail by the Court.” — Section 479(2), BNSS 2023

This sub-section has no equivalent in Section 436A CrPC and has been criticised on the ground that multi-offence FIRs are the norm rather than the exception in Indian policing, so that the exception risks swallowing the rule. [6][7] Whatever its merits as policy, it is the law, and it has been applied: in *K. Ramakrishna v. Assistant Director, Directorate of Enforcement* (Crl. P. No. 9930/2024, decided 23 November 2024) the High Court of Karnataka held that Section 479(1) is “subject to Section 479(2)” and that the provisions “have to be read conjointly”, refusing the benefit to an accused facing multiple cases. [10][11]

Consequently the engine treats 479(2) as a **hard gate evaluated before any threshold arithmetic is reported**, not as a footnote. A system that computes the fraction correctly but ignores 479(2) will confidently tell a large share of real

undertrials that they are eligible when the statute says they are not — the single most damaging failure mode available to a tool of this kind.

2.3 Section 479(3) — the automation hook

“The Superintendent of jail, where the accused person is detained, on completion of one-half or one-third of the period mentioned in sub-section (1), as the case may be, shall forthwith make an application in writing to the Court to proceed under sub-section (1) for the release of such person on bail.” — **Section 479(3), BNSS 2023**

This is the provision that turns the project from an advisory calculator into a workflow tool. The statute already requires a document to be produced by a named officer at a computable moment. Auto-drafting that application — pre-populated, provision-cited and ready for the Superintendent’s signature — is squarely inside the problem statement, requires no exercise of judicial discretion, and is directly aligned with the MHA advisory of 24 October 2024 which directs Jail Superintendents to prefer such applications on completion of the prescribed periods. [15]

2.4 The Satender Kumar Antil categories

In *Satender Kumar Antil v. Central Bureau of Investigation* (judgment of 11 July 2022) the Supreme Court restated that bail is the rule and jail the exception, and structured bail practice into four categories. [3][4][5] These map almost one-to-one onto a rules engine and form the backbone of the eligibility logic:

Category	Scope	Effect in the engine
A	Offences punishable with imprisonment of 7 years or less, not falling in Categories B or D	Ordinary summons route; strong presumption toward release; s.479 arithmetic applied directly
B	Offences punishable with death, life imprisonment, or imprisonment exceeding 7 years	Death/life offences excluded from s.479(1) by the main clause; otherwise merits-based, flagged for counsel
C	Offences under special Acts with stringent bail provisions — NDPS s.37, PMLA s.45, UAPA s.43-D(5), Companies Act s.212(6), POCSO	Statute-specific bail bar raised as a blocking condition; s.479 benefit reported only with the special-statute test attached
D	Economic offences not covered by special Acts	Merits-based; seriousness and quantum surfaced as facts, never scored

On the interaction between Section 479 and special statutes, the Supreme Court in *Badshah Majid Malik v. Directorate of Enforcement* (Crl. A. No. 4258/2024, 27 December 2024) held that Section 479(1) applies to prosecutions under the PMLA. [12] The same decision is reported as treating the provision as available to undertrials already in custody before 1 July 2024. Because the precise contours of retrospective operation are still being worked out through the 1382 Prisons proceedings and executive circulars, the engine will treat retrospectivity as a **configurable, cited assumption surfaced to the user** rather than a silent constant. [13][14]

3. Objectives

- **Charge-sheet parsing and cross-regime mapping.** Extract the offences and sections charged from unstructured charge sheets and FIRs and map them across IPC 1860, BNS 2023, BNSS 2023, BSA 2023 and the relevant special statutes, explicitly handling transitional cases in which pre- and post-1 July 2024 provisions appear together.
- **Complete Section 479 implementation.** Compute custody undergone against the one-half and one-third thresholds and apply the death/life exclusion, all three provisos, and the Section 479(2) multiple-case bar as explicit, individually testable gates.
- **Antil-structured eligibility logic.** Organise the rules engine around Categories A–D, including the stringent bail bars under NDPS s.37, PMLA s.45, UAPA s.43-D(5) and Companies Act s.212(6).
- **Section 479(3) application generation.** Auto-draft the Superintendent’s written application to the Court, and generate candidate lists in the format Undertrial Review Committees can act on.
- **Grounded precedent retrieval.** Retrieve judicial pronouncements through a RAG pipeline that returns citation, paragraph number and verbatim extract — never a generated summary of a holding.

- **Transparent, reproducible reporting.** Produce a section-wise report stamped with the statute version, the inputs used, a timestamp and a named human-reviewer field, so that any output can be reproduced against the law as it stood on a given date.
- **Accessible delivery.** Ship a multilingual web interface for the realistic access channels — DLSA legal-aid clinics, jail terminals, paralegal volunteers’ devices and family members — rather than claiming direct access by prisoners, who do not have internet access in custody.

4. Proposed Methodology

The system is organised as four layers. The design principle throughout is that the language model extracts and retrieves; it never adjudicates. Every legally operative conclusion is produced by deterministic code that can be read, tested and cited.

Layer A — Legal Knowledge Ingestion & Vectorization

Bare acts, special statutes and judgments are segmented using a hierarchy-aware strategy that respects the structure of statutory text (Act → Chapter → Section → sub-section → proviso) rather than fixed-size windows, since a proviso separated from its parent sub-section is worse than useless. Chunks are embedded with sentence-transformer models and indexed in FAISS. Because flat dense similarity over statutory text retrieves poorly — a query about Section 480 will happily return Section 479 — retrieval is hybrid: BM25 lexical matching combined with dense vectors, followed by cross-encoder reranking. Every chunk carries a statute version and in-force date so that outputs remain reproducible as the law changes.

Layer B — AI-Driven Charge Parsing & Section Extraction

A small domain-adapted language model — Phi-3 Mini, fine-tuned with LoRA/PEFT — extracts offences, section numbers, cognizability and compoundability from unstructured input. Parameter-efficient fine-tuning is chosen deliberately: the realistic deployment target is a district legal-services office or a prison network, not a GPU cluster, so a model that runs on modest on-premise hardware is a requirement rather than a compromise. Training and evaluation data are drawn from public sources — Indian Kanoon, eSCR, the NJDG judgment corpus, and the publicly released HLDC corpus of Uttar Pradesh district-court documents [22][23] — with named-entity redaction applied before any record is used. Charge sheets are not public documents and none will be scraped; realistic synthetic charge sheets, reviewed for legal plausibility, are used for parser evaluation.

Layer C — Structured Reasoning & Eligibility Engine

A deterministic rules engine sits on top of the extracted data and performs every legally operative step:

- cross-references each identified offence against a structured statutory-penalty database keyed to statute version;
- computes custody undergone, distinguishing date of arrest from date of first remand, since the two differ and the difference changes the arithmetic;
- applies the Section 479 thresholds and every gate in §2 above, including 479(2);
- enumerates procedural prerequisites — surety bond, personal bond, fine, identity verification — and flags cases suitable for release on personal bond in lieu of surety, since a person who cannot furnish surety stays in custody notwithstanding an order of bail [29];
- emits a section-wise rationale in the form *offence* → *section* → *maximum sentence* → *custody undergone* → *fraction served* → *applicable threshold* → *status*.

Two design rules constrain this layer. First, **discretionary factors are never scored**. Flight risk and the likelihood of influencing witnesses or tampering with evidence are matters of judicial discretion exercised on the record of a specific case; reducing them to weights produces a number that looks objective and is not, and — given that undertrial populations skew toward marginalised communities — creates a disparate-impact surface with no legal benefit. They are surfaced as an unweighted checklist which the report reproduces verbatim for a human to consider. Second, the engine is **asymmetric by construction**. A false “eligible” will be caught by the court; a false “ineligible” silently keeps a person in custody and nobody appeals a machine’s silence. The system therefore has no “ineligible” output state. It reports either “*statutory entitlement established*” or “*no statutory entitlement identified on these inputs — human review required*”, together with everything it checked and every gate that fired.

Layer D — RAG-Powered Precedent Retrieval

For discretion-heavy or ambiguous matters, a LangChain-based RAG pipeline retrieves relevant pronouncements from the vector store. Output is constrained to citation, paragraph number and verbatim extract; the system does not generate headnotes, because a synthesised summary of a holding is a hallucinated citation waiting to be relied on in court. Retrieval quality is measured with RAGAs — faithfulness, answer relevancy and context precision — reported continuously so that regressions are visible.

5. Technology Stack

Category	Technologies
Languages & Frameworks	Python; LangChain; PyTorch; Hugging Face Transformers; PEFT/LoRA; FAISS; Sentence-Transformers; rank_bm25
Backend	FastAPI (RESTful services); PostgreSQL for structured statutory-penalty and case data; MongoDB for document artefacts
Frontend	React responsive dashboard; multilingual UI with a separate translation/localisation layer
NLP & AI	Phi-3 Mini (LoRA/PEFT fine-tuned) for extraction; hierarchy-aware statutory chunking; hybrid BM25 + dense retrieval with cross-encoder reranking
Reasoning	Deterministic Python rules engine driven by a versioned, human-readable decision table (YAML/JSON) — no model in the decision path
Evaluation	RAGAs for retrieval quality; a hand-labelled golden test set with a separate correctness harness for the rules engine
Governance	Statute-version stamping; immutable audit log of inputs, rule firings and outputs; role-based access; on-premise deployment option

6. Evaluation Plan

Retrieval quality and legal correctness are different claims and are measured separately. RAGAs tells us whether the generated text is faithful to the retrieved context; it says nothing about whether the eligibility conclusion is right. Conflating the two would not survive scrutiny.

- **Rules-engine correctness.** A hand-built **golden test set of 150–200 cases** — drawn from reported judgments and realistic synthetic fact patterns — each annotated with a legally verified expected output, including deliberate coverage of 479(2) multiple-case scenarios, death/life exclusions, first-time-offender one-third cases, special-statute bars and transitional IPC/BNS cases. Reported as per-gate accuracy plus overall exact-match, with all failures published.
- **Extraction quality.** Precision, recall and F1 on offence and section extraction against a manually annotated held-out set.
- **Retrieval quality.** RAGAs faithfulness, answer relevancy and context precision, plus Recall@k and MRR against a set of queries with known correct statutory targets.
- **Operational metrics.** End-to-end latency per case, abstention rate, and the proportion of cases routed to human review.

The published literature supports this caution. The IBPS study of Indian bail prediction reports a best accuracy of 0.79 on a corpus of 150,430 High Court bail judgments using a fine-tuned Phi-4 model, and — instructively — found that naïvely adding retrieval to the baseline *reduced* accuracy from 0.47 to 0.33. [24] Retrieval augmentation is not free. Measuring it, rather than assuming it helps, is a design requirement.

7. Related Work and How This Project Differs

System / Work	What it does	Relationship to Bail Reckoner
HLDC (Findings of ACL 2022) [22][23]	912,568 Hindi district-court documents; 176,849 bail documents; binary bail-prediction task; best reported ~80% accuracy / 0.78 F1	Predicts the outcome a judge reached. Bail Reckoner computes a statutory entitlement. HLDC is used here as a public corpus for extraction training, not as an outcome model.
IBPS — Indian Bail Prediction System [24]	150,430 High Court bail judgments across five High Courts; Phi-4 + QLoRA; outcome prediction with explanation	Same distinction. Its finding that unstructured RAG degraded baseline accuracy directly informs this project’s hybrid-retrieval and evaluation design.
IL-TUR benchmark [25]	Benchmark suite for Indian legal text understanding and reasoning	A source of evaluation methodology for the extraction layer.
ICJS [28]	Inter-operable Criminal Justice System linking police, prosecution, courts, prisons, forensics	Moves records. Bail Reckoner is designed to consume ICJS-shaped data and return an entitlement computation — an integration target, not a competitor.
e-Prisons / BOMS	Prison management and bail-order management portals	The natural host for the s.479(3) workflow; the engine is designed to plug in behind it.
FASTER [27]	Supreme Court system for electronically transmitting bail and release orders to jails	Solves order delivery after bail is granted. Bail Reckoner addresses the stage before an application is ever filed.
UTRCs	District-level Undertrial Review Committees that identify prisoners eligible for release	The primary institutional consumer of the system’s output.

This clarifies a claim that should be scoped carefully rather than asserted broadly. It is *not* correct to say that no digital tool exists in this space — several national systems do. What is correct, and defensible, is that **the existing systems transmit records and orders; none of them computes bail entitlement under Section 479**. That reasoning layer is the contribution.

8. Expected Outcomes

- **Reproducible eligibility reports** — section-wise, provision-cited, stamped with statute version, inputs, timestamp and a named reviewer field.
- **Auto-drafted Section 479(3) applications** for jail Superintendents, and UTRC-ready candidate lists.
- **First-time-offender identification** — systematic surfacing of the one-third route under the first proviso, which remains under-invoked.
- **Verbatim precedent extracts** with citation and paragraph, suitable for direct use in a bail application.
- **A plug-and-play REST API** alignable with ICJS and e-Prisons deployments.
- **Two separately reported reliability figures** — rules-engine accuracy against the golden set, and RAGAs retrieval faithfulness. No accuracy figure will be claimed without a labelled set behind it.
- **Measurable social impact** — a shortened interval between the day an undertrial becomes statutorily entitled to release and the day a competent authority acts on it.

9. Responsible-AI Position, Governance and Data Protection

- **Facilitator, never decision-maker.** The system produces computations and citations for a human to act on. This is stated in the UI, in every generated report header, and in the API contract.

- **No scoring of discretion.** Flight risk and witness influence are checklist items reproduced verbatim, never weights. This also removes the principal fairness-audit surface, which matters because undertrial populations skew toward marginalised communities.
- **Asymmetric error design.** The system cannot output “ineligible”; the negative state is “no entitlement identified — human review required”.
- **Reproducibility.** Every output is stamped with the statute version and in-force date it was computed against, so a report can be re-derived months later.
- **Auditability.** An immutable log records inputs, which rules fired, retrieved sources and the output, with a named human reviewer field on every report.
- **Data protection.** Charge-sheet material is sensitive personal data. The design position under the Digital Personal Data Protection Act, 2023 [30] is data minimisation, purpose limitation, defined retention, redaction before any model training, and an on-premise deployment option so that identifiable data need never leave the institution holding it.
- **Scoped deployment claim.** The target is a credible pilot with one District Legal Services Authority and one district jail supporting UTRC workflow — not a national impact claim.

10. Known Limitations

- **Input quality is the binding constraint.** Charge sheets are frequently scanned, partly handwritten and in regional languages. OCR and Indic-language handling are non-trivial and are an explicit workstream rather than an assumption.
- **Phi-3 Mini is weak on Indic languages.** The multilingual objective is therefore met by a separate translation and localisation layer, not by the extraction model; this is stated rather than glossed.
- **The retrospectivity question is unsettled.** The scope of Section 479’s application to persons in custody before 1 July 2024 continues to evolve through the 1382 Prisons proceedings and executive circulars, and is therefore configurable and cited rather than hard-coded.
- **Eligibility is not release.** Prisoners remain in custody after bail is granted because surety or bond cannot be furnished. The system flags personal-bond suitability and legal-aid pathways, but cannot itself close that gap.
- **Statutory text must be verified against the official gazette.** Provisions quoted in this abstract were taken from bare-act reproductions and cross-checked across sources; the decision table will be encoded only from the official text.

11. Work Plan

Member	Primary responsibility	Key deliverable
Vallamalla Abhishek Prakash	Legal decision table (s.479 in full + Antil categories + special-statute bars); golden test set design; integration; documentation	Versioned decision table + labelled golden set
Revuru Arya	Layer A — statutory corpus ingestion, hierarchy-aware chunking, hybrid BM25 + dense retrieval, FAISS, statute versioning	Retrieval service with Recall@k report
Modhumpally Arvind	Layer B — charge-sheet parsing, Phi-3 Mini LoRA/PEFT fine-tuning, data sourcing and redaction	Extraction model + P/R/F1 report
Boru Vimala	Layer C & frontend — deterministic rules engine, report and s.479(3) application generation, React dashboard, audit log	Rules engine + generated application output
Shared	Layer D RAG pipeline, RAGAs evaluation harness, golden-set labelling and review	Evaluation report

12. References

All sources below were retrieved and checked in August 2026. Statutory text quoted in §2 was taken from bare-act reproductions [1][2] and cross-checked against secondary commentary [6][7][8][9]; it must be verified against the official gazette text before being encoded.

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